

MASSIMILIANO PARENTE

Department of Astronomy
University of Florida
211 Bryant Space Sciences Center,
Gainesville, FL 32611 USA
☎ +1 352 709 6038
✉ parente.m@ufl.edu
birth 10 Dec 1996, Lecce (IT)



RESEARCH INTERESTS

- Galaxy formation and evolution with hydrodynamic simulations and semi-analytic models
- Models for dust formation and evolution within galaxies
- Far-Infrared continuum and lines emission of galaxies
- Large scale structure impact on galaxy evolution

RESEARCH EXPERIENCE AND EDUCATION

- 2025-present **Postdoctoral Researcher**
University of Florida, Gainesville, FL (USA)
Dust and PAHs in galaxy evolution simulations – PI: Desika Narayanan
- 2024-2025 **Postdoctoral Researcher**
INAF, Trieste (IT)
Sub-mm emission lines in high- z simulated galaxies – PI: Umberto Maio
- 2020-2024 **PhD in Astrophysics and Cosmology**
SISSA (International School for Advanced Studies), Trieste (IT)
Evaluation: Summa cum laude
Thesis: Dust in hydrodynamic and semi-analytic galaxy evolution simulations
Supervisors: Gian Luigi Granato, Cinthia Ragone-Figueroa, Andrea Lapi
- 2018-2020 **Master of Science in Theoretical Physics**
Sapienza, Università di Roma, Roma (IT)
Evaluation: Summa cum laude
Thesis: The spectral properties and observability of dusty galaxies in the epoch of reionization
Supervisors: Raffaella Schneider, Luca Graziani
- 2020 **Excellence program**
Sapienza, Università di Roma, Roma (IT)
Numerical simulations on the evolution of the first metal free stars
Supervisor: Oscar Straniero
- 2015-2018 **Bachelor of Science in Physics**
Sapienza, Università di Roma, Roma (IT)
Evaluation: Summa cum laude
Dissertation on the red giant branch in galactic globular clusters

Supervisor: Oscar Straniero

2017-2018 **Excellence program**

Sapienza, Università di Roma, Roma (IT)

Extracurricular insights on the theory of number, electromagnetism and stellar evolution

2010-2015 **High school leaving qualification in scientific studies**

IISS Don Tonino Bello, Copertino (IT)

100/100 cum laude

PUBLICATIONS

- 2026 **Parente M.**, Narayanan D., Torrey P. OJA submitted. arXiv:2604.06314, "Dust and Grain Size Evolution in Galaxy Simulations: What Matters and What Does Not"
- 2026 **Parente M.**, Salvestrini, F., Granato, G. L., Narayanan, D., Tripodi, R., Bianchi, S., Bischetti, M., Feruglio, C., Fiore, F., Silva, L. A&A accepted. arXiv:2603.04505, "The Drivers of Cosmic Dust Temperature Evolution"
- 2026 Narayanan D., Torrey P., **Parente M.**, et al. OJA submitted, arXiv:2606.20809 "The Lifecycle and Emission Properties of PAHs in Cosmological Hydrodynamic Galaxy Formation Simulations"
- 2026 Venditti A., Graziani, L., Schneider, R., Bromm, V., ..., **Parente M.**, et al. 2026ApJ, 1005, 226V, "Catching the Nebular Needle in a Polluted Haystack: Line-emission Signatures from Population III-forming Pockets around Massive Galaxies at the End of Reionization"
- 2026 **Parente M.**, Bischetti M., Maio U., Salvestrini F., Feruglio C., Granato G. L., Ragone-Figueroa C., Tripodi R., De Breuck C., Ferkinhoff C., Tornatore L. A&A, 705, A39. doi:10.1051/0004-6361/202557033, "High- z [OI] emission lines: COLDSIM simulations and ALMA observations"
- 2025 Tripodi R., Bradac M., D'Eugenio F., ..., **Parente M.**, et al., arXiv:2507.20684 "A deep dive down the broad-line region: permitted OI, CaII and FeII emission in an AGN Little Red Dot at $z = 5.3$ "
- 2025 **Parente M.**, arXiv:2504.10585 "Modeling Dust in Galaxy Evolution Simulations"
- 2025 **Parente M.**, Ragone-Figueroa C., Granato G.L, Silva, L., Coenda, V., Martínez, H. J., Muriel, H., Lapi, A., A&A 697, 231, doi:10.1051/0004-6361/202452320, "The dust sub-mm emission of Green Valley galaxies"
- 2024 Nadolny, J., Michałowski, M.; **Parente, M.**; Solar, M.; Nowaczyk, P.; Ryzhov, O.; Leśniewska, A., A&A 693, id.L6, 7 pp., doi:10.1051/0004-6361/202452794, "Star formation rate surface density main sequence evolution. Insights from a semi-analytic simulation since $z = 12$ "
- 2024 Traina, A., Magnelli, B., Gruppioni, C., Delvecchio, I., **Parente, M.**, Calura, F., Bisigello, L., Feltre, A., Pozzi, F., Vallini, L., 2024, A&A 690, id.A84, 10 pp., doi:10.1051/0004-6361/202451113, "A³COSMOS: Dust mass function and dust mass density at $0.5 < z < 6$ "
- 2024 Ragone-Figueroa C., Granato G.L, **Parente M.**, et al., 2024, A&A 691, id.A200, 20 pp., doi:10.1051/0004-6361/202451344, "Intertwined Formation of H_2 , Dust, and Stars in Cosmological Simulations"

- 2024 Nadolny J., Michałowski M. J., **Parente M.**, *et al.*, 2024, A&A 689, id.A210, 13 pp., doi:10.1051/0004-6361/202449839, "The fate of the interstellar medium in early-type galaxies. IV. The impact of stellar feedback, mergers, and black holes on the cold ISM of simulated galaxies"
- 2024 **Parente M.**, Ragone-Figueroa C., Lopez P., Martinez H.J., Ruiz A.N., Ceccarelli L., Coenda V., Rodriguez F., Granato G.L., Lapi A., van de Weygaer R., 2024, ApJ, 966, Issue 2, p. 17, doi:10.3847/1538-4357/ad320e, "Star formation and dust in the cosmic web"
- 2023 **Parente M.**, Ragone-Figueroa C., Granato G.L., Lapi A., 2023, MNRAS, 521, 6105. doi:10.1093/mnras/stad907, "The $z \lesssim 1$ drop of cosmic dust abundance in a semi-analytic framework"
- 2022 **Parente M.**, Ragone-Figueroa C., Granato G.L., Borgani S., Murante G., Valentini M., Bressan A., Lapi A., 2022, MNRAS, 515, 2053. doi:10.1093/mnras/stac1913, "Dust evolution with MUPPI in cosmological volumes"

SEMINARS AND CONFERENCES

- 2026 **Exploring the Aromatic Universe in the JWST era**, London, Canada
Speaker, "What PAHs reveal about galaxy evolution"
6-10 Jul 2026
- 2026 **Lunch Talk**, ESO, Garching
Speaker, "Dust and PAHs: a window on galaxy evolution"
30 Jun 2026
- 2026 **Seminar**, University of Gent
Speaker, "Dust and PAHs: a window on galaxy evolution"
25 Jun 2026
- 2026 **Seminar**, INAF, Bologna
Speaker, "Dust and PAHs: a window on galaxy evolution"
9 Jun 2026
- 2026 **The Drivers of Galaxy Evolution through Cosmic Time**, Florence
Speaker, "Dust and PAHs in galaxy evolution"
8-12 Jun 2026
- 2026 **From Dust Till Dawn**, Leiden
Speaker, "Modeling dust sizes in galaxy simulations"
23-27 Feb 2026
- 2026 **UF colloquium**, Gainesville
Speaker, "Dust *et al.* in galaxy evolution simulations"
29 Jan 2026
- 2025 **L-Galaxies workshop 2025**, Berlin (*online*)
Speaker, "Dust emission in L-Galaxies"
15-17 Oct 2025
- 2025 **Peering through an opaque ISM 2025**, Ferrara
Speaker, "dust *et al.* in galaxy evolution simulations: predictions and physical relevance"
1-5 Sept 2025
- 2025 **Hydrosim collaboration meeting 2025**, Trieste

- Speaker, "High redshift [OI] emission in ColdSIM simulations"
28-31 July 2025
- 2025 **IR Fine-Structure Lines Workshop**, Winona
Speaker, "High redshift [OI] emission from cosmological simulations"
10-13 June 2025
- 2025 **Seminar**, University of Florida
Speaker, "Modeling the IR Side of Galaxies: dusty simulations and more"
04 June 2025
- 2024 **Evolution of Dust and Gas throughout Cosmic Time**, Hiroshima
Speaker, "The $z \lesssim 1$ drop of cosmic dust abundance"
09 - 13 Dec 2024
- 2024 **Hydrosim collaboration meeting 2024**, Ljubiana
Speaker, "Clues on SMBH growth and galaxy quenching from a large scale perspective"
29 Jul - 1 Aug 2024
- 2024 **The origin and evolution of SMBHs**, Sexten
Speaker, "From voids to clusters: the impact of SMBHs of the star formation in the cosmic web"
15-19 Jul 2024
- 2024 **Bridging the models and observations of galaxies' dust in the JWST era**, IFPU, Trieste
Speaker, "Modelling dust in galaxy evolution simulations"
22-26 Apr 2024
- 2024 **L-Galaxies seminars**, online
Speaker, "The $z \lesssim 1$ drop of cosmic dust abundance "
12 Mar 2024
- 2024 **L-Galaxies Workshop 2024**, Donostia
Speaker, "Galaxy evolution within the cosmic web: clues to SMBH growth in L-Galaxies2020"
22-24 Jan 2024
- 2023 **4th Regional Extragalactic Astronomy Meeting: 60 Years Of The Sersic Law**, Córdoba
Speaker, "From voids to clusters: star formation and dust in the cosmic web"
4-7 Dec 2023
- 2023 **Seminar**, IATE, Córdoba
Speaker, "Dust in galaxy evolution: from grains properties to the cosmic web"
01 Dec 2023
- 2023 **IFPU Focus Week Galactic Archeology**, IFPU, Trieste
Speaker, "Modeling Dust in galaxy evolution simulations"
9-12 Oct 2023
- 2023 **Origin and Fate of Dust in Our Universe**, Gothenburg
Speaker, "The $z < 1$ drop of cosmic dust abundance"
25-29 Sep 2023
- 2023 **Cosmology 2023 in Miramare**, Trieste

- Speaker, "On the environmental dependence of galaxy properties in dusty semi-analytical models"
28 Aug - 02 Sep 2023
- 2023 **Hydrosim collaboration meeting 2023**, Sexten
Speaker, "To Sim or To SAM? Dust evolution in hydrodynamic simulations and semi-analytic models"
10-15 July 2023
- 2023 **PhD Seminar**, INAF, Trieste
Speaker, "Dust in galaxy evolution: a semi-analytic approach"
15 June 2023
- 2023 **Astronomy Seminar**, ICC, Durham
Speaker, "Dust in galaxy evolution: a semi-analytic approach"
6 June 2023
- 2022 **Hydrosim collaboration meeting 2022**, Trieste
Speaker, "Dust and H₂ in MUPPI"
11-13 Sep 2022
- 2022 **Seminar**, IATE, Córdoba (online)
Speaker, "Dust evolution in cosmological simulations"
6 May 2022
- 2021 **Hydrosim collaboration meeting 2021**, online
Speaker, "Dust evolution in cosmological simulations with MUPPI"
17 Nov 2021

ACADEMIC EXPERIENCE

- Referee for *Astronomy & Astrophysics*
 - Referee for *The Astrophysical Journal*
 - Referee for *Monthly Notices of the Royal Astronomical Society*
- 2023 Member of the LOC of the "Cosmology 2023 in Miramare" conference
- 2023 Webmaster of the *Astrophysics and Cosmology group* webpage
- 2020 Tutor of Physics course for pharmacy students
Sapienza, Università di Roma, Roma (IT)

AWARDS

- 2025 'Premio Tacchini' award for the best PhD thesis in astrophysics in Italy
Awarded by *Italian Astronomical Society*
- 2021 'Outstanding Graduate' award for the academic year 2019/2020
Awarded by *Sapienza Università di Roma* and *Fondazione Roma Sapienza*
- 2018 INFN prize for best students in nuclear physics
Visit at CERN laboratories

SKILLS & LANGUAGES

Programming C, C++, Python, IDL, L^AT_EX, Unix/Linux/macOS
and softwares

English Fluent
Spanish Basic
Italian Mother Tongue

ACADEMIC REFERENCES

- **Dr. Gian Luigi Granato**
INAF – Trieste (Italy)
e-mail: gianluigi.granato@inaf.it
- **Dr. Cinthia Ragone-Figueroa**
IATE – Córdoba (Argentina)
e-mail: cinthia.ragone@unc.edu.ar
- **Prof. Cedric Lacey**
Durham University – Durham (UK)
e-mail: cedric.lacey@durham.ac.uk
- **Prof. Andrea Lapi**
SISSA – Trieste
e-mail: lapi@sissa.it